

Marketing Attribution & Budget Intelligence

□ Core prediction pipeline

1 Value Proposition

Last-touch attribution misreads what drives conversions. We replace it with **Markov Chain attribution** + a conversion predictor — smarter budget calls, same spend.

2 Data Sources

- **586,737** interactions, **240,108** customers
- 5 channels: FB, Search, Instagram, Video, Display
- ID, timestamp, channel, type, conversion

3 Prediction Task

- **Attribution:** Markov removal-effect — not a prediction
- **Propensity:** converts within 7 days? (binary)

4 Decisions

- Reallocate budget across 5 channels
- Who to remarket to

5 Making Predictions

- **Attribution:** batch, on data refresh
- **Propensity:** real-time, FastAPI /predict

6 Offline Evaluation

- Chosen via validation PR-AUC
- Test: **ROC-AUC 0.648** · **PR-AUC 0.065** · **Brier 0.035**
- Customer-level split, zero overlap

7 Data Collection

- Real ad spend
- Demographics, device, time-of-day
- A/B holdout → causal incrementality

8 Features

- Touchpoint count — **~35%** importance
- Unique channels · per-channel visits
- First / last channel seen

9 Building Models

- **Logistic Regression** (final) vs. **XGBoost**
- PR-AUC over reweighting — 4.4% conversion
- 7-day horizon; 78,025 censored rows excluded

10 Live Eval & Monitoring

- CI suite: 15 tests, every change
- Health check: DB, model, metrics
- *Next:* retraining + drift monitoring